

Drinking Water — Specification (IS 10500)

Scope

IS 10500 prescribes the acceptable and permissible limits for the physical, chemical, bacteriological and organoleptic parameters of drinking water. The standard is used across India for testing water quality in domestic supply, packaged drinking water plants and community water systems. Organoleptic parameters cover colour, odour and taste; physical parameters include turbidity and total dissolved solids. Certain limits are marked 'relaxable' with the consent of the authority if no alternative source is available.

Key Requirements

Bacteriological requirement: coliform bacteria shall not be detectable in 100 ml of the sample. Chemical limits cover pH, total hardness, chlorides, sulphates and heavy metals such as lead, arsenic, cadmium and mercury. The standard defines acceptable limits (for normal supply) and permissible limits (in the absence of an alternative source). Test methods referenced include the standard methods for examination of water and wastewater.

Testing & Compliance

Sampling shall be carried out from the point of consumption, not only from the distribution network. The full analytical schedule covers more than fifty parameters. Continuous supply systems require residual free chlorine verification where chlorination is practised. Results are compared with Table 1 (acceptable) and Table 2 (permissible in the absence of alternative source).